

so increase speed to keep rate constant

(1)

- 5 (a) into (plane of) paper/downwards 1 [1]
- (b) (i) the centripetal force = mv^2/r 1
 $mv^2/r = Bqv$ hence $q/m = v/r B$ (some algebra essential) 1 [2]
- (ii) $q/m = (8.2 \times 10^6)/(23 \times 10^{-2} \times 0.74)$ 1
 $= 4.82 \times 10^7 \text{ C kg}^{-1}$ 1 [2]
- (c) (i) mass = $(1.6 \times 10^{-19})/(4.82 \times 10^7 \times 1.66 \times 10^{-27})$ 1
 $= 2u$ 1 [2]
- (ii) proton + neutron 1 [1]

- 6 (a) (i) either probability of decay or $dN/dt = \lambda N$ OR $\lambda = (1/N) dN/dt$ 1